Washroom-related Google Maps reviews of 47 highway restaurants on the Bengaluru to Big Bay corridor

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Mayur S P and Apeksha S

Atria University, Bengaluru, India.

Correspondence: mayurshivaprem@gmail.com, apekshaammu09@gmail.com

Abstract

Washroom hygiene shapes where travelers stop on Indian highways, but it is rarely recorded in structured form. We present 5,162 Google Maps reviews of 47 restaurants rated above 3.5 along the roughly 65 km route from Phoenix Mall of Asia, Bengaluru, to the Big Bay pitstop on National Highway 44. Reviews were collected mainly through Apify, with SerpApi for targeted washroom searches, merged on the Google review identifier, and span April 2016 to October 2026. A rule-based matcher identified 372 reviews that discuss a washroom, and only the washroom passage of each was kept. Every passage carries an aspect-based sentiment label and a score from a pretrained DeBERTa model. The data can serve as a benchmark for aspect-level sentiment on Indian English reviews, for mapping roadside sanitation, and for stop-recommendation tools for drivers.

Background & Summary

Drivers on long highway trips often pick a stop by whether its washroom is usable, yet listing ratings blend food, service and price into one star score. Public aspect-based sentiment benchmarks such as SemEval-2014 Task 4 cover restaurant aspects like food and service, not sanitation, and contain no Indian highway venues[1].

This dataset isolates what reviewers say about washrooms at every well-rated restaurant on one busy corridor, Bellary Road and National Highway 44 north of Bengaluru. It keeps the full merged review history for context and adds a sentiment label for each washroom passage. The data can be reused to train or test aspect-level sentiment models, to compare stops before a trip, or as a template for surveying other routes.

Methods

Restaurant selection. Google Places text searches were run for each stretch of the route: Jakkur, Yelahanka, Bagalur Cross, Airport Road, Devanahalli, Chikkaballapur and Peresandra. Restaurants on the highway, or within about 2 km of it, with a Google rating above 3.5 were kept, giving 47 venues. Airport-terminal outlets and venues on the Nandi Hills road were excluded because they need a detour of about 6 km or more.

Identifiers. Each Google Place ID was decoded into the two 64-bit integers it encodes, written as the hexadecimal Maps `data_id` (`0x…:0x…`) that review APIs accept.

Collection. The pipeline ran in four steps:

1. The newest reviews of every venue (target 100) were downloaded with the Apify Google Maps Reviews Scraper (compass/Google-Maps-Reviews-Scraper). Apify supplied 4,666 of the 5,162 reviews.
2. SerpApi's Google Maps Reviews engine added earlier newest-first pages and four in-review searches per venue (washroom, restroom, toilet, bathroom). It supplied 496 reviews that Apify did not return, including 335 of the 372 washroom reviews.
3. All sources were merged on Google's `review_id`, so each review appears once with a `sources` field recording where it came from.
4. Dates were normalised to ISO 8601 and reviewer names were dropped from the tabular output.

Washroom extraction. A review counts as washroom-related when its text contains washroom, restroom, toilet, bathroom, loo, lavatory, WC, urinal or hand-wash area. Matching is case-insensitive and whole-word, tolerates common misspellings, and ignores figurative uses such as "bathroom-themed". Only the sentences about the washroom, plus a short follow-on sentence, form the snippet, so food comments do not leak into the sentiment.

Review photos. For six venues, the images attached to reviews were downloaded from each review's own `images` field. Each file was checked to be a real image from its first bytes, named after its review, and stored once even when several reviews share it.

Sentiment. Each snippet was scored locally with `yangheng/deberta-v3-base-absa-v1.1`[2][3][4], using the washroom word the reviewer wrote as the aspect. The score is P(positive) minus P(negative), from −1 to +1, averaged over passages when a review has several.

Ethics. Only publicly posted reviews were used. Reviewer names and profile links are excluded from the tabular files.

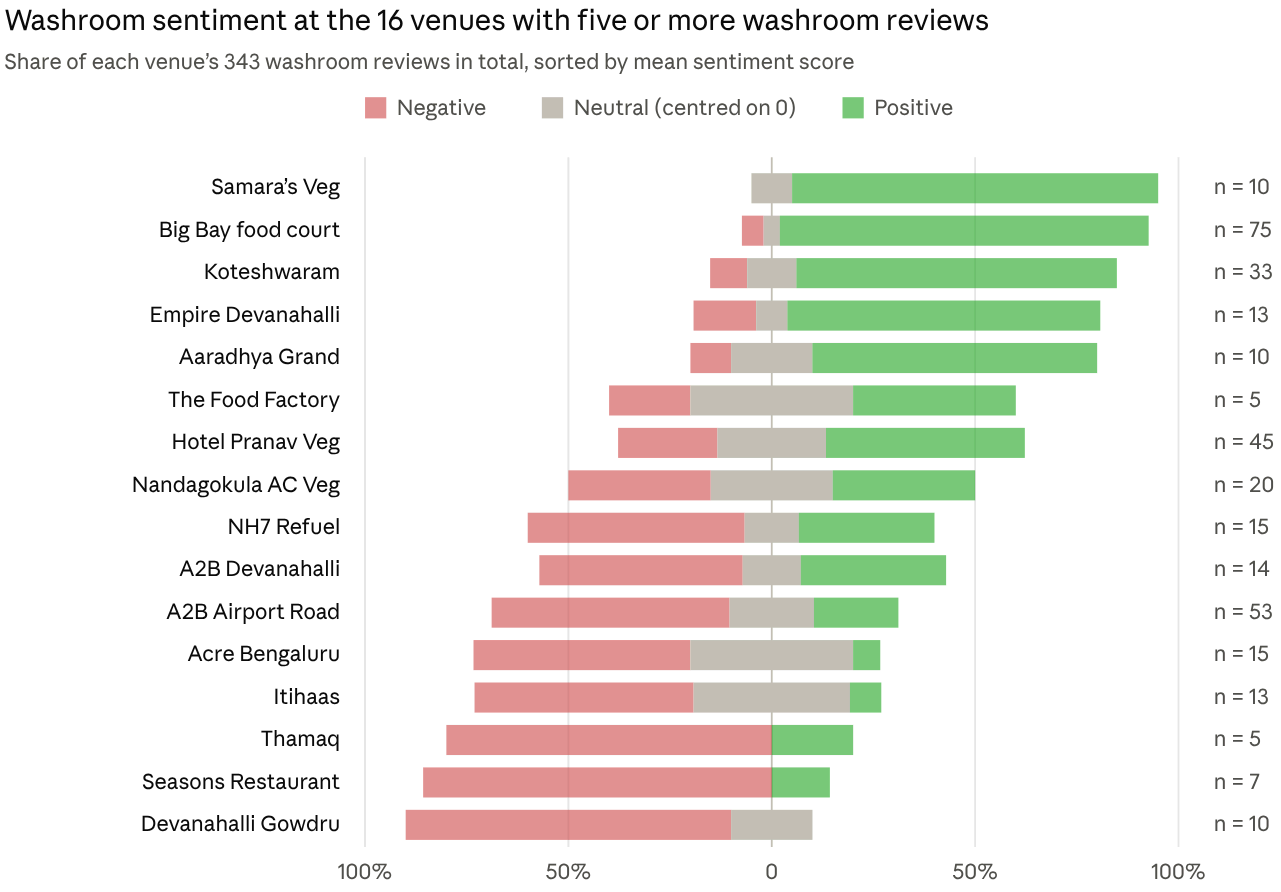
Data Records

The dataset is held in the Pitstop GitHub repository[5] under `data/`. All tables are UTF-8 CSV and per-venue records are JSON.

File	One row per	Key fields
`raw_data/restaurants_route_above_3.5.csv`	venue (47)	name, location, `data_id`, rating, review count, cuisine, on-highway or detour
`processed/all_reviews.csv`	review (5,162)	`data_id`, `review_id`, `iso_date`, rating, `sources`, `matched_by`, language, translated, text, link
`processed/reviews/<data_id>_reviews.json`	venue (47)	the merged reviews with per-source counts
`processed/washroom_mentions_scored.csv`	washroom review (372)	`washroom_snippet`, `passage_count`, star rating, `sentiment`, `sentiment_score`, `confidence`, `p_negative`, `p_neutral`, `p_positive`, `needs_review`
`processed/washroom_summary.csv`	venue (47)	sample size, reviews with text, `washroom_share`, washroom reviews, latest mention date
`processed/washroom_sentiment_summary.csv`	venue (47)	positive, neutral and negative counts, mean and recent (last 2 years) score, `low_data`

Raw API responses (API keys redacted) are kept under `raw_data/serpapi/` and `raw_data/apify/` for provenance. The unscored washroom mentions are also in `processed/washroom_mentions.csv`, and the reviews from each source are kept separately in `output/`, `output/washroom/` and `output/apify/`. Review photos downloaded for six venues (168 image files) are stored in `extracted_images/<data_id>/`, and each one is linked to its review in `output/<data_id>_reviews.json`.

Data Overview



data/processed/washroom_sentiment_summary.csv · the 16 venues with low_data = False

Figure 1. Washroom evidence is uneven. Four venues (Big Bay, A2B Airport Road, Hotel Pranav Veg and Koteshwaram) hold 206 of the 372 washroom reviews, and 31 venues have fewer than five. Among well-covered venues, Big Bay and Samara's Veg are about 90% positive, while A2B Airport Road, Seasons and Devanahalli Gowdru are mostly negative.

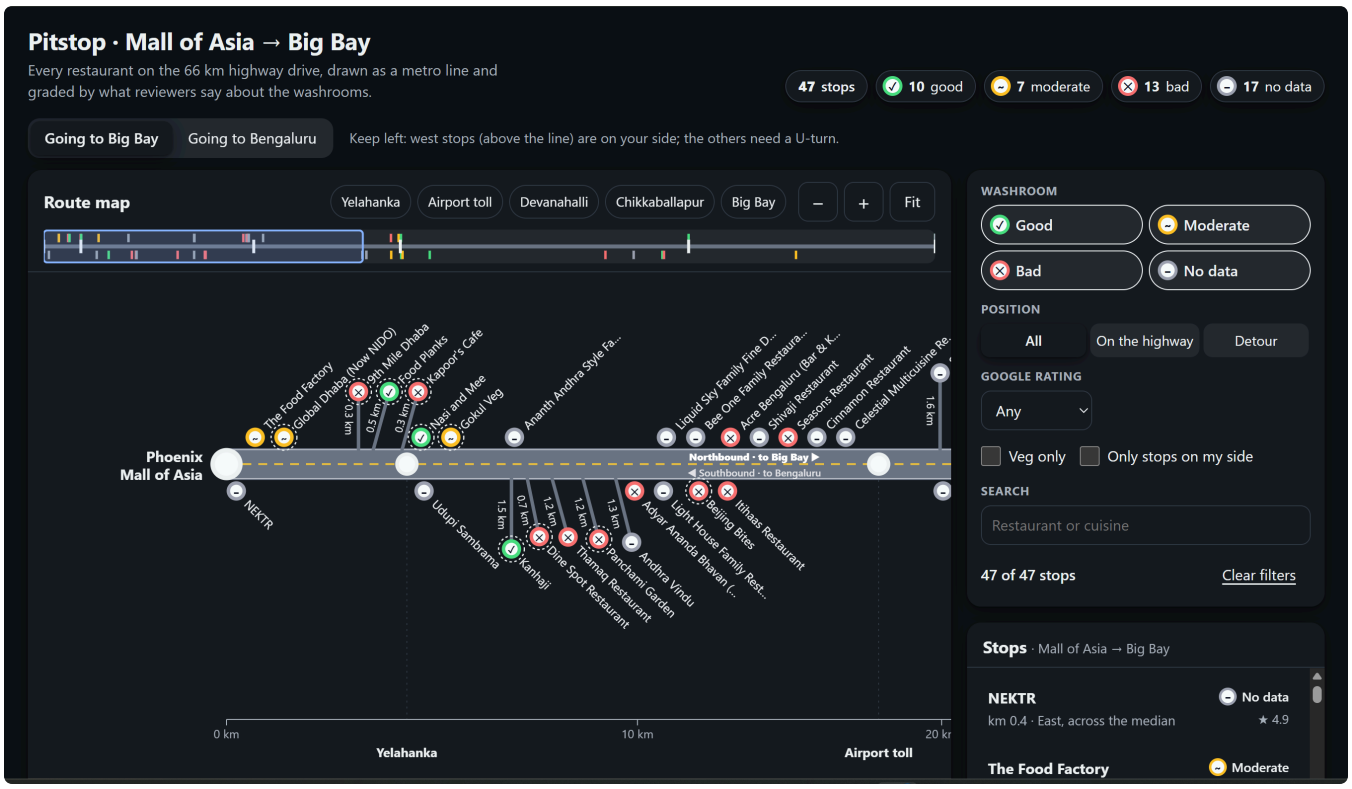


Figure 2. Route view of the dataset. Each of the 47 restaurants is a station on the 66 km road route from Phoenix Mall of Asia to Big Bay. West-side restaurants sit above the two-lane road and east-side ones below, and restaurants off the highway hang on a perpendicular spur labelled with their distance from the road. Stations are coloured by mean washroom sentiment score: good at 0.3 or above (10 venues), moderate between −0.3 and 0.3 (7), bad below −0.3 (13), and no washroom reviews (17). Dashed rings mark venues with fewer than five washroom reviews.

Technical Validation

- **Identifiers.** All 47 decoded `data_id` values returned reviews for the intended venue, and every venue appears in the merged file.
- **Uniqueness.** The 5,162 (venue, `review_id`) pairs are all distinct, and none is missing a `review_id`.
- **Matcher.** An offline suite of 297 mocked tests covers the pipeline, and all of them pass at commit 49039ce. It includes false-positive cases ("rested", "toiletries", "Waterloo", "there was room") and snippet boundaries.
- **Sentiment.** Model scores correlate with the review's star rating (Pearson r = 0.63, n = 372). Mean ratings are 2.43 stars for negative, 3.26 for neutral and 4.45 for positive washroom passages. The mismatches that remain are mostly reviews whose

washroom opinion differs from their overall verdict. Sixteen passages with confidence below 0.6, or with conflicting labels across passages, are flagged `needs_review`.

- **Coverage.** Google's in-review search under-returns matches. For Big Bay, the four searches found 22 of the 57 washroom reviews present in its newest unfiltered reviews. Unfiltered download plus local matching was therefore used as the primary source.

Usage Notes

Washroom counts are lower bounds. Only 37 of the 2,629 newest reviews with text (1.4%) mention a washroom, and 17 venues have none. Thirty-one venues have fewer than five washroom reviews and are marked `low_data`, so their mean scores should not be compared with those of well-covered venues. Fifty-two reviews are Google machine translations, mainly from Hindi and Kannada, and carry `translated = true`. Sentiment can be re-scored with another model by reading `washroom_snippet` directly. Re-collection needs a SerpApi key and an Apify token in `sources/.env`. Review text remains subject to Google Maps terms of service.

Data Availability

All files described in Data Records are available in the Pitstop repository at github.com/apeksha0463/Pitstop under `data/`, as of commit `49039ce` (30 October 2026)[5]. [Before submission: Scientific Data asks for the data to be in a recognised repository with a DOI. Archive this commit on Zenodo or figshare, add the DOI here, and choose a licence, since the repository has none yet.]

Code Availability

All code is in the same repository under `scripts/` and is written in Python 3.10+. The main dependencies are requests 2.31+, python-dotenv 1.0+, sentencepiece, protobuf, PyTorch 2.0+ and Hugging Face Transformers 4.40+[4]. The workflow runs `collect_latest_reviews.py`, `collect_washroom_reviews.py`, `collect_apify_reviews.py`, `merge_reviews.py`, `extract_washroom_mentions.py` and `score_washroom_sentiment.py` in that order. `extract_review_images.py` and `restaurant_photos.py` download review and listing photos. The test suite runs offline with `python -m pytest scripts/tests`.

References

1. Pontiki, M. et al. SemEval-2014 Task 4: aspect based sentiment analysis. In *Proc. 8th International Workshop on Semantic Evaluation* 27–35 (2014). https://doi.org/10.3115/v1/S14-2004
2. He, P., Gao, J. & Chen, W. DeBERTaV3: improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. Preprint at https://arxiv.org/abs/2111.09543 (2021).
3. Yang, H., Zhang, C. & Li, K. PyABSA: a modularized framework for reproducible aspect-based sentiment analysis. Preprint at https://arxiv.org/abs/2208.01368 (2022).
4. Wolf, T. et al. Transformers: state-of-the-art natural language processing. In *Proc. 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* 38–45 (2020). https://doi.org/10.18653/v1/2020.emnlp-demos.6

Author Contributions

Mayur S P. and Apeksha S. contributed equally to this work. Both selected the route and restaurants, built the review collection, merging and washroom-extraction pipeline, ran the sentiment analysis and technical validation, and wrote and revised the manuscript.

Competing Interests

The authors declare no competing interests.

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